Raghavendra Pedada

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Portfolio: raghavendrap.vercel.app

Professional Summary

CS undergraduate (IoT, Cybersecurity & Blockchain) experienced in full-stack development, AI-powered systems, RAG pipelines, and edge AI. Delivered 4+ end-to-end projects using Python, FastAPI, React, Docker, GitHub Actions, Raspberry Pi, and TFLite. Passionate about scalable software, on-device AI inference, and performance-oriented engineering.

Education

B.Tech, Computer Science & Engineering *Expected: 2027*
Specialization: IoT, Cybersecurity & Blockchain
Alva's Institute of Engineering and Technology (VTU), Mangaluru CGPA: 8.03

Intermediate (Class XII) — Narayana Junior College, Andhra Pradesh 07/2023 78%

Secondary School (Class X) — Sri Chaithanya Techno School, Andhra Pradesh 07/2021 99%

Technical Skills

Languages	Python, Java, C++, C, JavaScript, SQL
Frameworks	FastAPI, React, Vite
Web Tech	HTML5, CSS3, JavaScript, REST APIs
AI/ML & Edge	Prompt Engineering, RAG, LLMs, TensorFlow Lite, MediaPipe, OpenCV
Embedded	Raspberry Pi, ESP32, Arduino, MQTT, Sensors
Core CS	Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks
Databases	SQLite, MySQL
Libraries	SQLAlchemy, Chart.js, Framer Motion, Axios
Tools	Git, GitHub, Docker, GitHub Actions, Linux, VS Code

Experience

Cybersecurity Micro-Intern *2024*
Tata Consultancy Services (TCS) — Virtual Programme

- Completed programme covering network security, threat modelling, and vulnerability assessment.
- Evaluated IoT device security postures and proposed mitigation strategies for common attack vectors.

Projects

HashPilot – AI-Powered Computational Benchmark Platform *FastAPI • React • SQLite • Docker • AI*
GitHub: GitHub

- Developed a full-stack AI-powered computational benchmarking platform to evaluate Proof-of-Work strategies including Sequential, Random, MultiThread, and MultiProcess implementations.
- Built a FastAPI backend exposing REST APIs for benchmark execution, analytics, benchmark history, AI-powered strategy recommendations, and system information collection.
- Designed an interactive React dashboard with benchmark visualization, analytics, performance charts, winner detection, and responsive user experience.
- Integrated SQLite for persistent benchmark storage, Docker for containerization, and GitHub Actions for automated testing and CI/CD workflows.

Cosmic RAG System *Python • RAG • LLMs*
GitHub: GitHub

- Built a Retrieval-Augmented Generation pipeline for intelligent document question answering.
- Integrated semantic retrieval with Large Language Models to generate context-aware responses.
- Designed a modular architecture supporting scalable AI-powered document intelligence applications.

AI Tool Calling Assistant *Python • APIs*
GitHub: GitHub

- Developed an AI assistant capable of invoking external tools through structured function calling.
- Integrated API-driven workflows to automate multi-step tasks and improve productivity.
- Applied modular software engineering principles to build scalable and maintainable AI workflows.

Edge-AI Smart Mirror *Final Year Project • Raspberry Pi 4B • TFLite • MediaPipe • OpenCV*

- Built a real-time, on-device health and wellness monitoring system on Raspberry Pi 4B with zero cloud dependency.
- Implemented posture detection, mood analysis, and rPPG-based vitals estimation using MediaPipe and TFLite inference pipelines.
- Designed modular OpenCV-based vision architecture enabling concurrent multi-metric tracking at low latency on edge hardware.

IoT Health Monitoring System *ESP32 • MQTT • Python • Sensors*

- Designed an IoT-based health monitoring system using ESP32 microcontrollers and biometric sensors for real-time vital data acquisition.
- Published sensor telemetry over MQTT to a Python backend for processing, logging, and threshold-based alerting.
- Demonstrated end-to-end embedded-to-cloud pipeline bridging hardware and software layers.

Certifications

- **IBM/Coursera** — CI/CD with OpenShift: GitHub Actions & Tekton Pipelines
- **TCS** — Cybersecurity Micro-Internship (Virtual)

Links

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